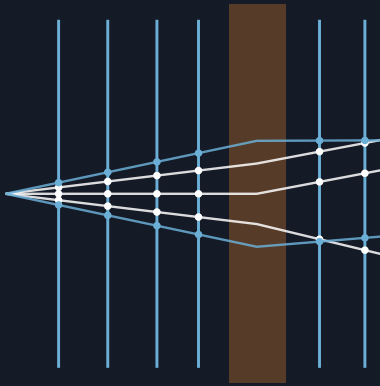
MUON BACKGROUND, DESIGN SPACE AND RECONSTRUCTION, LIVE

A veto for SHiP

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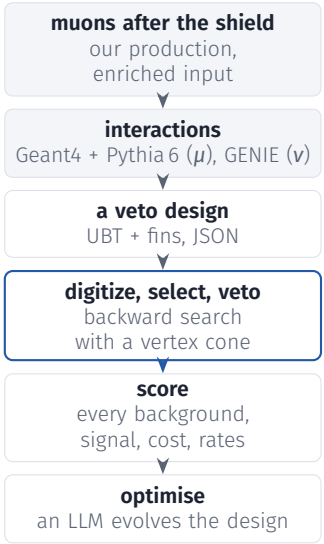
How to read this talk

Every plot with a slider is computed while you watch, by the code of the study (`veto/lib/veto`):

- muons after the shield from our own production (the enriched input), and the older samples
- interactions of muons and neutrinos after the shield (fast-veto-sim)
- digitization, random hits, the selection and the veto
- designs scored on every background source, by hand and by an LLM-driven evolution

Slow widgets show a spinner and fill in; 3D views rotate and zoom. The spill is **1 s** of $5 \cdot 10^{13}$ POT, weights are muons per spill (= Hz); 15 years = $1.2 \cdot 10^{7}$ spills.

> **The standard**: **100 signal events** (dark-matter decays passing the selection) in 15 years. **FP** / **TN**: how many of the 100 the veto throws away / keeps. **TP** / **FN**: background events in 15 years vetoed / let through.



PART 1

The problem

Muons from the target reach the detector. Two of them can fake a decay in the vessel, and one muon or neutrino interacting after the shield makes a real vertex. The veto has to see them coming in, without killing the signal.

SHiP and its muon background





Target at $z = 0$, **muon shield** (TRY6) to 30 m.

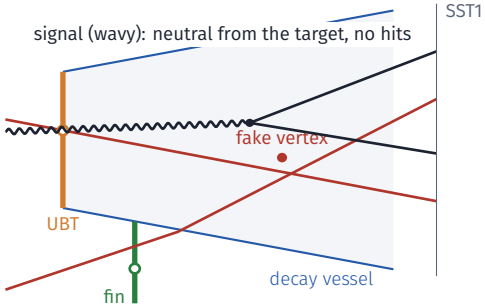
Decay vessel $z = 30.6$–80.6 m, 1×2.7 m entrance widening to 4×6 m.

UBT on the entrance window, **SST1** at 82 m, spectrometer magnet and timing detector behind.

Cavern wall at $x = -4.56$ m, floor at $y = -3.36$ m.

Muons per type (slider): bounced into SST1 · SST1 $p > 1$, timing · SST1 other · through an SBT wall · through the UBT window · bounced in the rock · straight past · stop in the rock · stop in the shield.

What the veto has to do



Two muons from the same spill, within ±200 ps, can form a vertex inside the vessel that points back to the target: a **fake signal**.

Each came in from outside: through the UBT window, the side walls, or after bouncing off the cavern. A hit where its track came from **explains** it.

muons per spill (simulated)	$1.8 \cdot 10^{10}$
crossing $z = 30.6$ m	$3.5 \cdot 10^{9}$
in the SST1 acceptance	$3 \cdot 10^{4}$
fake-vertex pairs, 15 years	**36** before any veto

> Veto every fake vertex, but do not veto the signal with **random hits** of the 10^{9}–10^{10} other muons.

Three kinds of background

	combinatorial	**muon interactions**	**neutrino interactions**
origin	two unrelated muons within ±200 ps	one muon interacts in the helium, the walls, the air or a veto plane	one neutrino interacts in the same materials
vertex	fake (a coincidence)	real	real
can the veto see it?	yes: each muon came in from outside	yes in principle: the parent crossed the UBT or a fin, but it was bent at the vertex	no: nothing charged comes in
handled by	the veto	the veto with a **vertex cone**	the **selection** only

The first studies only had the first column. The other two are larger by orders of magnitude before any cut; they changed the selection, the veto and the objective.

PART 2

One particle at a time

Before any statistics: what a single muon does after the shield, where it crosses the detector, how it bounces, scatters and interacts.

One muon at a time

live:mu_type

live:mu3d

live:mu_index

15 typical muons per type (full run, drawn by weight); straight lines between the stored states. · z = 30.6 m, ■ z = 82 m or stop, × bounce.

ONE PARTICLE AT A TIME

Bouncing off the cavern

live:bounce_view

live:bounce_map

live:bounce_ev

Grey dashed: straight line from the 30.6 m state; orange dotted: backward line from the 82 m state. They meet within 4 cm (median) at the bounce point, on the $-x$ wall or the floor. Right: all reconstructed bounce points trace the cavern.

One muon at a time: what it crosses

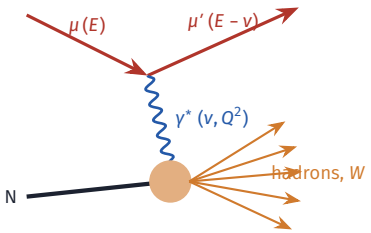
live:sim_mu_one3d

live:sim_mu_pick

live:sim_mu_design

Muons that do interact somewhere. Blue: helium chord · red: PVC (exact chord) · grey: air · dots: planes. $P = n_{\text{area}}\,\sigma(E)$, the chance to interact there.

Muon--nucleus interactions



$W^2 = M^2 + 2M\nu - Q^2$

The muon exchanges a virtual photon with a nucleus: the muon goes on, **bent by a few mrad**, and hadrons come out (pions, kaons, protons, K^0_S, Λ that decay later).

Geant4 (muonNuclear): the soft part, resonances and nuclear break-up. **Pythia 6**: the hard part, $W > 2$ GeV. **crossover** (our choice): Geant4 below an energy transfer $v_x(E)$, Pythia above.

E_μ [GeV]	10	100	300
σ per nucleon [μb]	8.7	19.8	26.2
of which Pythia	1 %	11 %	15 %

Library: 50 energies 0.5--500 GeV, 20 k events per energy and material.

One interaction at a time

live:sim_int3d

live:sim_i_E

live:sim_i_ev

live:sim_i_zoom

live:sim_i_gen

Library events: beam along z; length $\propto \log p$; thick: the scattered muon; dashed: neutral; : decay point. E: 3, 10, 30, 100, 300 GeV.

One applied interaction at a time

live:sim_ai3d

live:sim_v_ev

live:sim_v_tgt

live:sim_v_kind

Black: the parent muon (A, K, B); dotted grey: the parent neutrino. Products straight to $z = 82$ m. "→ SST1": charged, $p > 1$ GeV, in $|x| < 2$, $|y| < 3$ m.

Decays: from where, into what

live:sim_decay3d

live:sim_dk_part

live:sim_dk_p

K^0_S, hyperons, π^0, η are decayed in the library (main channels, decay point from $c\tau$): a K^0_S made in the wall can decay inside the vessel into $\pi^+\pi^-$, a real two-track vertex.

Multiple scattering: one muon at a time

live:sim_msc_traj

live:sim_ms_traj_p

60 muons on the axis; red: a hard single scatter. 1.5, 3, 10, 30, 100 GeV.

PART 3

Data and simulation

Our own production of the muons after the shield, an input enriched at high energy, and the forced interactions of muons and neutrinos on top. Simulated once; every design is then evaluated in seconds.

Two simulations of the same muons

	20 001 old runs	**the full run** (`Full_dataset_z_3060_8200.pkl`)
muons	only those ending in the SST1 acceptance, ≈ 1 300 per run, 26.5 M rows	all 506 M simulated, one row each, 32 GB
states	start, z = 30.6 m, z = 82 m	z = 30.6 m and z = 82 m, or where the muon stopped
good for	pair background: ≈ $2 \cdot 10^5$ fake-vertex candidates	everything that misses the spectrometer: UBT and fin rates, veto-only muons
read with	a column store, 78 s to build	memory-mapped in place, a pass in ≈ 1 min

- The full run is a **rerun of seed 42**: same input muons, same states at 30.6 m; old file 42 holds its SST1 muons. Combined, it is not a 20 002nd run.
- `evt_n` = input-muon index in both; weights belong to the input muon (identical in both).
- Muons that bounce off the cavern differ between the two: bouncing is random from run to run.

Where the muons go

live:where_bar

70 % stop in the shield; 26 M range out in the rock behind the cavern wall.

125 M pass the vessel to z = 82 m outside the spectrometer: harmless themselves, but they light up any detector placed in their way.

Only ≈ 2 000 per run matter directly: they enter the vessel or reach SST1.

> The veto sees 10^9–10^{10} muons per spill to catch ≈ 10^4. Accidental hits are the real enemy.

Where the muons go, in Hz

live:where_hz

The same muons as rates: muons per spill of $5 \cdot 10^{13}$ POT over the 1 s spill.

≈3.5 GHz cross z = 30.6 m and 2.7 GHz pass the vessel to z = 82 m.

≈29 kHz reach the SST1 acceptance (the three SST1 bars together), ≈27 kHz enter the vessel without reaching it.

> A veto detector sees GHz in total; the muons it must catch arrive at tens of kHz.

Our own muon production

`bin/generate_muon_study.py`: G. Frisella's shield simulation (TRY6, GPU propagation) imported unchanged; every muon taken to z = 30.6 m and 82 m and classified on the GPU (torch, identical to the CPU version, 2 × 151 M muons checked).

One seed = one spill: the same input muons, new random numbers. ≈ 95 s per seed on one GPU.

Every category is kept up to 15 M rows, also the muons that stop in the shield. Storage is *fill until the budget*: type t (rate R_t per seed) is kept by the first

$$n_t = \max\left(\lceil 15\,\mathrm{M}/R_t \rceil,\ 20\right)$$

seeds, each a random fraction re-weighted to the seed's total. SST1 and veto-only muons: all of them, in every seed.

type	per seed	seeds kept
0 SST1, bounced	299	all
1 SST1, selected	301	all
2 SST1, other	723	all
3 veto only (walls, window)	975	all
4 bounced in the rock	575 k	27
5 straight past the vessel	125 M	20
6 stops at 30.6–82 m	26 M	20
7 stops in the shield	355 M	20

Guglielmo's study files had 1 500 picks per category, overlapping categories and a window cut too narrow for the UBT: the reason for our own production.

20 000 seeds on farm-gpu3, 4 GPUs (PR #7).

The datasets

	built from	rows	size	used for
alpha (14 Oct)	20 001 old runs (SST1 types) + the full run (everything else)	57 M	5.8 GB	first results, first evolution
gamma (17 Oct)	SST1 types from alpha; types 3–7 from 1 715 seeds of our production	88 M	8.9 GB	interactions, cone
delta (19 Oct)	everything from 484 seeds of the **enriched** production (input with the Pythia 6 tail)	111 M	11 GB	**the end-to-end results**
epsilon (21 Oct)	all 6 000 seeds of the enriched production	552 M	55 GB	next results
new storage (22 Oct)	800 seeds of a new storage rule (see below)	73 M	7 GB	common types, rate maps

Same columns in all three: states at 30.6 and 82 m, start state, bounce point, walls crossed, type, seed, `evt_n` (the input muon). `weight` is always muons per spill: a seed's rows of a type are re-weighted to its total and divided by the number of seeds that kept the type.

Rows are not independent: every seed re-simulates the same input muons. What counts is the number of **independent input muons**, $n_{\text{eff}} = (\sum w)^2 / \sum w^2$ with the weights summed per `evt_n` first.

> More seeds of the same input do not help where a few heavy input muons dominate.

The enriched input: a Pythia 6 tail



In the 506 M input muons, everything above 150 GeV from minimum bias is a few rows of weight 768.75 each: in gamma **10 input muons carry 32 %** of the selected SST1 muons (n_{eff} = 55).

`fast-veto-sim` replaces them by **Pythia 6.4.28** muons (first collision, Geant4 cascade, π/K forced to decay, Mo+W target), normalised to the same flux above 150 GeV: 601 M rows.

Selected SST1: n_{eff} 55 → **18 200**. The harder spectrum also gives **40–55 % more** SST1 muons.

> Veto-only muons (walls, window) are unchanged at ≈ 12 k: not the high-energy tail.

Enough statistics everywhere?







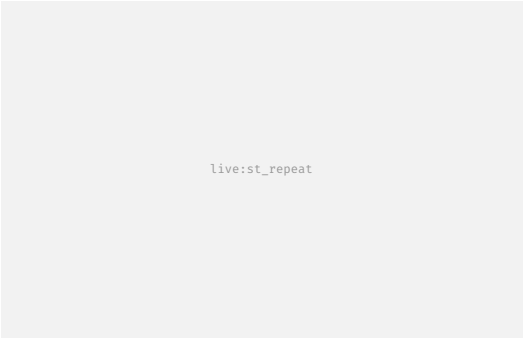
gamma delta; dotted: 10

A probe, not a design: the UBT at full height, 42 fins every 2.5 m and 5 floor planes tilted by 30°; per 25 cm cell the independent input muons crossing it.

Thin: the first metre from the vessel wall (median 11–14), the floor beyond 32 m (8–13); fins beyond 1 m and the UBT: 140–430.

More seeds: what they buy





Every seed replays the same 601 M input muons; only the randomness of the shield transport makes new ones reach a place. **Bounced** muons grow with the seeds (×12 from delta to epsilon); **SST1 selected** only ×1.5: it is limited by the input file. The common types (straight past, rock bounced, stops) are kept by the first 20–27 seeds only: more seeds add **nothing** there.

Left: delta's first 60--484 seeds, epsilon's 6 000. Right: probe cells, types kept completely (delta).

Where the storage goes

live:st_plane_maps

live:st_plane

live:st_view

Today: a **uniform** random 0.3 % of each common type, in the first seeds. The rows follow the muons: the hot lobe 1.5–3 m out is full, the cells near the wall and on the floor get a handful.

A storage rule that knows where muons cross

Per seed, each muon of the common types is kept with

$$p_i = \max\left(p_{\text{floor}},\ \min\left(1,\ \lambda\, w_i \max_{c \ni i} \frac{1}{V_c}\right)\right)$$

- c: probe cells (25 cm) on its path; V_c: muons per spill there, from the seed itself
- $\propto w$ in a cell: equal weights w_i / p_i; every cell the same number of rows
- p_{floor}: 500 per type and seed, shield stoppers too
- λ: 10 000 rows per seed; SST1 and veto-only stay complete
- rate maps: histograms of **every** muon

Right: emulated on delta's rows (Horvitz--Thompson), same 58 M rows over 6 000 seeds.



The new storage, tested

live:st_bands

delta (484 seeds) epsilon (6 000) new storage (800)

live:st_bands_metric

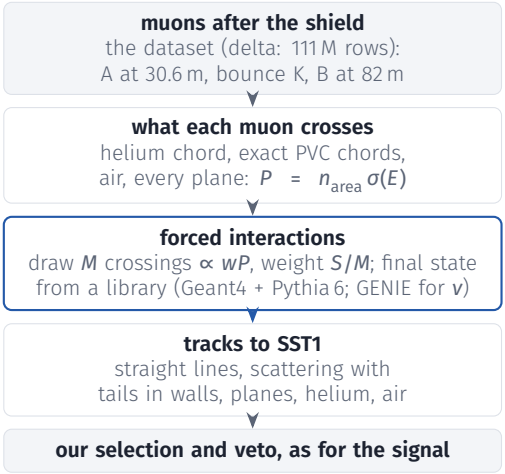
`generate_muon_study.py --storage alpha`: 800 seeds of the enriched input, 73 M rows (less than delta), 14 h on 3 GPUs.

Weakest cells (10th percentile) **23–80×** better than epsilon in every band: 110 next to the wall, 70 on the floor, 1 500 on the fins beyond 1 m. Cells below 10: 30–44 % → **0–1 %**.

Medians 8–150× higher too.

> Use it with epsilon: SST1 and veto-only types from epsilon's 6 000 seeds, everything else and the rate maps from here.

From the shield to SST1: the chain



Built in `fast-veto-sim` (Python + numpy at run time), used as it is.

step	time
muon interactions, one design (4 M)	60–80 s
cache: helium, PVC, air (11 M, delta)	89 s
neutrinos (4.2 M, GENIE library)	~90 s
one design, all backgrounds	**3--6 s**

Helium, PVC and air do not depend on the design: drawn once. Each design only needs its veto.

Generators: Geant4, Pythia 6 and GENIE

Libraries, built once per material and energy (fast-veto-sim):

- **muons**: Geant4 muonNuclear for the soft part (resonances, nuclear break-up), **Pythia 6** for the hard part ($W > 2$ GeV); our default *crossover*: Geant4 below an energy transfer $\nu_x(E)$, Pythia above. 50 energies 0.5–500 GeV, 20 k events per energy and material.
- **neutrinos**: **GENIE 3.06.02** (G18_02a, Pythia 6 hadronisation, charm and τ decays): ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$ on every nucleus of our materials, 1–500 GeV.
- short-lived products (K^0_S, hyperons, π^0, η) decayed in the library, decay point from $c\tau$.

Applying a library to our muons (or the neutrino flux):

1. every crossing of helium, PVC, air and each plane: $P = n_{\text{area}}\,\sigma(E)$;
2. per target, **force** M interactions drawn $\propto wP$, each of weight S/M, $S = \sum wP$: the weights add up to the true rate;
3. vertex uniform along the crossing; a library event at a neighbouring energy, rotated onto the direction, random azimuth;
4. the scattered muon continues from its **parent's own path**: the parent's real hits upstream are digitised.

> Helium, PVC and air do not depend on the design: drawn once and cached. A design's own planes are drawn per design.

Forced interactions

Real interactions are 10^{-6}–10^{-8} per crossing: far too rare to simulate muon by muon. So:

1. all crossings, $P = n_{\text{area}}\sigma(E)$; per target group g the total $S_g = \sum wP$ (interactions per spill);
2. draw M_g crossings with probability $\propto wP$ and **make them interact**, each with weight S_g/M_g: the weights add up to the true rate;
3. vertex uniform along the crossing;
4. final state: a library event at a neighbouring energy, rotated onto the muon's direction, random azimuth.

Every number later is $\sum$ weight $\times\, 1.2 \cdot 10^7$ spills (15 years); errors: bootstrap over input muons.

Muon interactions per 15 years:

helium	$1.5 \cdot 10^6$
PVC walls	$2.2 \cdot 10^7$
UBT (straw)	$3.2 \cdot 10^4$
one tile fin at 30.6 m	$\sim 10^{11}$
cavern air, 30.6–82 m	$1.4 \cdot 10^{12}$

Helium agrees with the full-Geant4 rate study (1.46 M).

> Most happen outside the vessel (air, fins): their products rarely make a vertex inside. The helium and the walls matter.

Cross sections

live:sim_xs_mu

live:sim_xs_split

live:sim_xs_mat

crossover: Geant4 below $\nu_x(E)$, Pythia above · w2: Geant4 below $W = 2$ GeV · geant4: Geant4 only. Pythia gives 1.4–1.7× more extra tracks: the generator systematic.

Grazing walls: a factor 4.4

live:sim_wall

live:sim_w_ang

The side walls lean outward by 30 mrad. Muons cross them at a few degrees or less, where the path in 0.3 mm of PVC is centimetres to metres (here: 40 mm on average).

Our old scattering capped it at 50× the thickness: **4.9 M** interactions in the walls per 15 years. Exact chords: **21.6 M**, as the full-Geant4 studies (18–19 M). Now in our scattering too.

Right: angle to the wall of muons crossing a side wall; red line: the old cap.

Where the interactions are

live:sim_vtx3d

live:sim_v_tgt

live:sim_v_kind

Light blue: helium vertices, red: wall vertices (samples); black: the chosen target.

Muon vertices in the helium follow the muons that enter the vessel; neutrino vertices fill it around the axis.

D6_straw; the helium, PVC and air rates are the same for every design.

Neutrinos

fast-veto-sim's own **GENIE 3.06.02** (G18_02a, Pythia 6 hadronisation, charm and τ decays): a library of ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$ on every nucleus of our materials, 1–500 GeV, applied to the neutrino flux through the same forced-interaction machinery.

No incoming charged particle: nothing for a veto plane to see. Only the selection can remove them.

One neutrino interaction at a time: Part 2, *One applied interaction at a time*, from: neutrinos.

Per 15 years (D6_straw; helium, PVC and air are design independent):

target	interactions	≥ 2 tracks in SST1
helium	5 290	1 360
PVC walls	3 690	374
air	5 690	881

After our selection (Part 5): **0.8** per 15 years, ≈ 0.5 after the veto's random hits: a floor for every design.

Multiple scattering: the tails

live:sim_msc_hist

live:sim_ms_p

live:sim_ms_q

Our scattering (now with fast-veto-sim's tails): Highland's Gaussian core over walls, planes, **helium and air** (1 m slices, on now), exact wall chords.

Tails: rare large single scatters. Above $2\,\theta_0$, a Poisson number per scatterer from Rutherford, $1/\theta^4$ up to the nuclear size; one parameter tuned to Geant4.

RMS and 99 % quantile at 82 m: within 5–8 % of Geant4 at 1.5–100 GeV; Gaussian alone: 99 % 25 % too small.

Fast chain vs full Geant4

live:sim_closure

2 M interactions in helium and PVC in full Geant4 (everything transported to 82 m) and the same draws through the fast chain. Charged tracks besides the muon, $p > 1$ GeV, in SST1:

fast vs full (helium)	≥ 1	≥ 2
Geant4 physics	−5 %	−9 %
crossover (ours)	×1.3	×1.6

Missing in the fast chain: photon conversions, secondary interactions. Our default (crossover) is the **conservative** side for two-track vertices.

PART 4

The design space

A UBT on the vessel entrance and thin fins reaching out from the side walls, left and right. Nothing on top or below.

UBT and fins















Sliders: UBT top · fins per side · gap · fin length · fin top [m]. Fin colour: $\log_{10}$ muons/m^2/spill, 2 (purple) to 9 (yellow). Black: drawn muons crossing nothing. Bars: geometric coverage.

Where the muons are

live:rate_map

live:map_z

live:map_side

Muon density on a fin plane at z, every muon of the dataset; u is the distance from the vessel wall. Side: 0 right ($-x$), 1 left ($+x$). Dotted: cavern wall and floor limit.

A **lobe** at $y \approx 0$, 1.5–3 m from the wall, reaches 10^9 muons/m^2 per spill: the muons the shield sweeps past the vessel. Close to the wall and away from $y = 0$ it is 3–5 orders of magnitude quieter.

> A fin pays for every muon crossing it. The muons it is there for cross it near the wall.

Seven designs to learn from

		area
D0	UBT on the entrance only	3 m^2
D1	taller UBT, one 4 m fin per side at 30.6 m	56 m^2
D2	4 fins per side every 15 m, 3 m long, $y \pm$ 3 m	145 m^2
D3	7 fins on the right up to the wall, 3 short on the left	209 m^2
D4	8 fins per side, 5 m, y to 6 m: the stress test	624 m^2
D5	D2 with 50 cm or 5 cm cells	145 m^2
D6	D2-like with real technologies: straw, fibre, pixel, tile, tile2	–
E1–E3	evolved: a minimal UBT and two short tile2 fin pairs (30.6, 36 m)	4 m^2

A design is a small JSON file: UBT edges and cell size; per fin its side, z, length, y range, cell size; detector efficiency and time resolution.

The loader applies the physical limits (floor, cavern wall, minimum UBT) and says what it changed.

All: 10 cm cells, ε = 0.99, σ_t = 0.5 ns, no tilt yet.

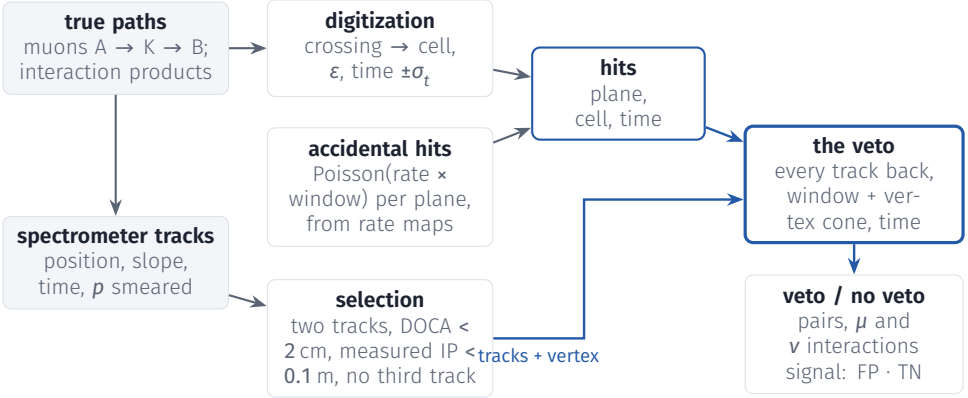
> In the RL environment fins are placed one after another along z with a minimum gap.

PART 5

Reconstruction

Events are short time windows around a spectrometer candidate. The selection keeps two-track vertices pointing at the target; every track is followed back to the veto planes, in a window that grows from the vertex.

The chain



Four samples, one chain: muon pairs, muon and neutrino interactions, and the signal (no own veto hits: only random hits veto it).

Events are time windows, not spills





cell size [cm]; 1 s spill

A fin at 30.6 m sees ~ 10^9 muons per second: simulating whole spills hit by hit is hopeless.

Instead, around each candidate: the own muons' hits, plus per plane

$$n_{\text{acc}} \sim \text{Poisson}\left(R_{\text{plane}} \cdot \frac{2W}{T_{\text{spill}}}\right)$$

random hits, placed according to the full-statistics rate map and digitized like real ones.

Blue box: a search window near the wall. What matters is the chance a random hit lands in it: rate density × box × window.

Orange squares: random hits of one ±1.5 ns window on the right fin of D2, at the chosen cell size.

The detector and spectrometer models

Veto planes (UBT and every fin, the same model)

- one layer of cells at fixed z (10 cm default)
- a crossing makes one hit with probability ε (0.99)
- hit = cell centre + time, σ_t = 0.5 ns, muons at c
- no direction, no energy, no noise, no dead time (yet)
- a muon's own crossing comes from its true path, bounce included

Spectrometer (track at z = 82 m)

- position ±1 mm, time ±0.1 ns
- slope ±0.1 mrad ⊕ 2 mrad·(1 GeV/p)
- momentum as true, used only to size the window

Direction comes only from the spectrometer. The UBT gives no usable direction; neither do the fins.

Samples

- combinatorial: opposite-charge muon pairs of one seed in the selection, weight $w_1 w_2 N_{\text{runs}} \cdot 2\Delta t / T_{\text{spill}}$, Δt = 200 ps: 197 k (alpha); 25 M (delta), subsampled ∝ weight to 400 k (total within 0.05 %)
- signal: long-lived neutrals from the target ($\theta \approx p_T/p$), decaying in the vessel into two tracks
- single SST1 muons (200 k) for per-muon efficiencies
- forced μ (11 M) and ν (7–11 M) interactions

The signal: from the target, not from Mars



A long-lived neutral from the **target**: p = 10–100 GeV (log-uniform), p_T ~ exp(1 GeV), so $\theta \approx p_T/p$.

Decays uniformly in z in the vessel, m = 1 GeV → two muon-mass tracks.

The pairs' selection: SST1, p > 1 GeV, timing, IP < 2.5 m. **37 %** pass; median daughter angle 22 mrad.

Orange: its tracks followed back to z = 30.6 m; grey: muon density.

> Two thirds cross the quiet UBT; the rest hit the fins next to the wall, where random hits veto them.

The selection

For **every** sample (signal, pairs, interactions):

1. two opposite-charge tracks, $p > 1$ GeV, in the SST1 and timing acceptance;
2. DOCA < 2 cm, vertex in the vessel;
3. **pointing**: the measured tracks (smeared at 82 m) give an IP at the target < 0.1 m;
4. **two tracks**: no other track in SST1 in the event.

A signal decay from the target points back (measured IP: median 2 cm, 99.9 % below 8.3 cm) and has two tracks; interactions have more tracks, and their muons were swept by the shield.

What survives the new cuts (3 + 4):

sample	before	after
signal	100 %	**99.98 %**
combinatorial pairs	36	2.5
muon interactions	$1.25 \cdot 10^5$	$2.0 \cdot 10^3$
neutrino interactions	481	**0.8**

Per 15 years, before the veto. Two-body signal model: decays with an invisible particle point back less well.

Algorithm A: follow every track back

For each spectrometer track:

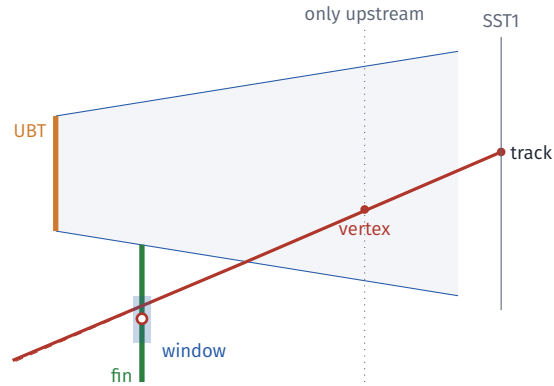
1. extrapolate backward in a straight line from $z = 82$ m (no field in the vessel)
2. on every veto plane **upstream of the vertex**: predicted cell (u, y) and time $t - \ell/c$
3. look for a hit within

$$|\Delta u|, |\Delta y| \le r_0 + \left(a + \frac{b}{p}\right) d + \tfrac{\text{cell}}{2}, \quad |\Delta t| \le n\,\sigma$$

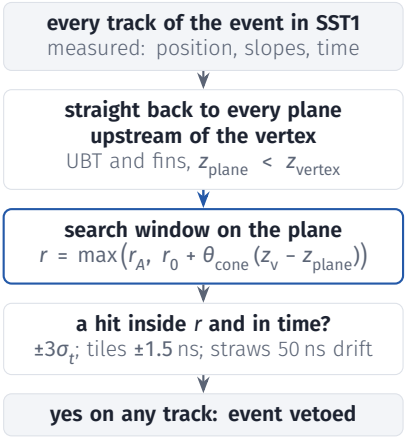
4. a hit found: the track is **explained**

A candidate is vetoed if either track is explained.

> No combinatorics: one binary search per (event, plane) and a scan of that plane's few hits.



The veto algorithm, step by step



Algorithm A (as before): $r_A = r_0 + (r_{slope} + r_{ms}/p)\,d + \frac{cell}{2}$, d the distance from SST1: $r_0 = 5$ cm, $r_{slope} = 0.5$ mrad, $r_{ms} = 6$ mrad GeV. Hits: the event's own (efficiency, cells, timing) and random ones from the rate maps.

New 1, every track: not only the two vertex tracks. The scattered muon of an interaction, followed back past the vertex, points at its parent's hit.

New 2, the vertex cone: the parent was bent at the vertex by a few mrad, so its hit is off the backward line by ≈ angle × distance *from the vertex*: the window grows from the vertex. A design parameter: wider catches more parents and more random hits (signal loss).

Single muons: no vertex, unchanged.

The spatial window

live:window_plot

live:r_ms

$r = r_0 + (a + b/p)\,d$ with $r_0 = 5$ cm, $a = 0.5$ mrad and b the slider, in mrad·GeV: the spectrometer's slope error grows with the lever arm d, multiple scattering with $1/p$.

Title: how many of the true crossings of straight SST1 muons (D2, 60 k muons) fall inside the window, computed now.

> For straight muons the prediction is within millimetres: the window can be tight. A bigger window only buys more random hits.

RECONSTRUCTION

The time window

live:time_plot

live:time_trade

live:sig_t

live:n_sig

The predicted time includes the time of flight from the plane to z = 82 m. The window is $\pm n\sigma$ with $\sigma = \sigma_t^{\text{veto}} \oplus \sigma_t^{\text{track}}$: wider loses fewer own hits, but lets in proportionally more random hits.

Right: signal loss scaled from D2's 1.3 % at 3σ and 0.5 ns, since random hits grow with the window.

Why the parent is missed, and the cone



live:iv_ev

live:iv_cone

Selected muon interactions (helium / walls), E3_robust, the heaviest 200. Black: the parent up to the vertex; red: the scattered muon; blue: products. Switch the cone and watch the windows grow.

Algorithm A on real events





design: 0 D0 · 1 D1 · 2 D2 · 3 D3 · 4 D4 · 5 D5 coarse · 6 D5 fine · 7 fibre · 8 pixel · 9 straw · 10 tile · 11 tile2 (fins) · 12 E1 · 13 E2 (LLM).

kind: 0 pair vetoed · 1 pair missed · 2 signal lost to a random hit · 3 bounced muon missed.

Solid: true path; dotted: backward prediction; white: own hits; diamond: the hit that explained the track; grey: random hits.

The same event from above



x–*z* view. Thick lines: UBT (orange) and fins (green). Blue bars: the search window on each plane the track is followed to. Same sliders as the previous slide.

What it costs

design	planes	hits / event	comparisons	µs / event
D0	1	0.1	40 k	0.5
D1	3	16	3.2 M	4.8
D2	9	39	4.0 M	12.5
D3	11	49	4.9 M	16.7
D4	17	90	7.2 M	33.3

196 736 background events; numpy, one core of a laptop.

A full evaluation of a design (196 k pairs, 100 k signal, 200 k single muons) takes **≈6 s** on one core, linear in hits per event.

Nothing to cap: algorithm A never combines hits.

The interaction backgrounds add 3–6 s: only their veto runs per design.

> Tens of seconds per design on the cluster, 8 in parallel: fast enough for an evolution loop.

PART 6

Results

One design through everything: the enriched dataset, every background source before and after the veto, the vertex cone, the signal, the cost.

The designs, and two iterations

live:e2e_design3d

live:e2e_design

E3_robust: from the first evolution runs (E2 + 10 cm margins; see Evolution): the minimal straw UBT and two short tile2 fin pairs at 30.6 and 36 m, 26 k channels, 0.40 MCHF. **D6_tile2**: the hand-made reference, tile2 fins every 15 m.

Iteration 1: E3's surviving muon interactions all have a parent that crossed *one* plane, the UBT or a 30.6 m fin, and left no hit: plane efficiency 0.999. Hence a **second station** where the parents enter:

- **E4_ubt2**: two UBT stations (+97 kCHF)
- **E4_entry2**: two stations on the UBT and both 30.6 m fins (+206 kCHF)

Iteration 2: the cone of E4_entry2, 0–20 mrad.

A second station (`"stations": 2`): an identical layer behind, efficiency $1 - 10^{-6}$, twice the channels and cost.

RESULTS

Every background, before and after the veto

live:e2e_sources

live:e2e_cone

live:e2e_design

Light: selected, before the veto; dark: after it. Per 15 years, enriched dataset, the full chain: scattering with tails, helium and air; measured IP < 0.1 m; at most two tracks; the cone of the design.

Muon interactions in the PVC walls (130–170) and in the veto planes (75–80) are all vetoed: their parents crossed the planes right before. In the **helium**, 4 300 are selected.

> Neutrinos: ≈0.9 selected, a third vetoed by random hits; ≈0.6 is a floor for every design.

Why the last muon interactions survive

The survivors of E3_robust at 10 mrad, one by one (the parent at z = 30.6 m):

- all from the **high-energy tail**: 130–300 GeV, the enriched input's own muons
- parents enter at $|x|$ = 0.43–0.54 m: through the UBT window or the $-x$ wall right behind the 30.6 m fin
- in truth every parent crossed a plane; the event simply has **no hit** from it: one plane, efficiency 0.999
- the scattered muon is reconstructed in every case; its kink is 1–2 mrad

$4\,300 \times 10^{-3} \approx 4$ per 15 years: exactly what is left once the cone recovers the kinks.

Two separate effects, two separate cures:

missed because	how often	cure
bend at the vertex	most (no cone)	vertex cone
no hit on its plane	10^{-3}	second station

> Redundancy is not about geometry here: the parents of interactions come in through the entrance, where one plane is the only chance.

RESULTS

Signal against background

live:e2e_trade

Each line one design, each point a cone (labels in mrad); all background sources together, per 15 years, against FP: of 100 signal events, how many random hits veto.

The cone costs signal through the random hits it lets in: cheap on short fins near the wall (E3, E4: FP 1 at 10 mrad), expensive on D6_tile2's busy fins (FP 6 at 10, 18 at 20 mrad).

> With a second station at the entrance the muon interactions go to zero at 17.5 mrad: what is left is the neutrino floor.

End-to-end figure of merit

			FN per 15 years						
design	cone	**FP**	comb.	μ helium	ν helium	**total**	MCHF	sens.	reward
E3_robust	0	0.06	$9 \cdot 10^{-5}$	76 ± 6	0.72	76 ± 6	0.40	0.14	−87.1
	10	0.92	$4 \cdot 10^{-5}$	3.1 ± 1.3	0.64	3.8 ± 1.3		0.47	−54.2
	20	3.60	$3 \cdot 10^{-5}$	1.6 ± 0.9	0.54	2.1 ± 0.9		0.54	−46.4
E4_ubt2	10	0.90	$4 \cdot 10^{-5}$	1.6 ± 0.9	0.67	2.2 ± 0.9	0.50	0.55	−46.0
	16	2.32	$4 \cdot 10^{-5}$	0.5 ± 0.5	0.63	1.2 ± 0.5		0.65	−36.2
E4_entry2	10	0.90	$2 \cdot 10^{-5}$	1.0 ± 0.7	0.67	1.7 ± 0.7	0.61	0.59	−41.9
	16	**2.35**	$1 \cdot 10^{-5}$	**0**	0.63	0.63 ± 0.06		**0.74**	−27.3
	20	3.78	$1 \cdot 10^{-5}$	0	0.61	0.61 ± 0.06		0.73	−27.9
D6_tile2	10	5.89	$3 \cdot 10^{-5}$	1.0 ± 0.7	0.43	1.5 ± 0.7	4.4	0.59	rates
	20	17.8	$1 \cdot 10^{-5}$	1.0 ± 0.7	0.37	1.4 ± 0.7		0.52	rates

FP: of 100 signal events. FN: per 15 years after selection and veto; μ interactions in air, PVC and the planes and ν in air and PVC: 0 in every row. Sensitivity and reward: see Optimisation. 100 k signal decays, 400 k pairs, 11–16 M μ + 7 M ν forced interactions; delta dataset, 484 seeds. D6_tile2: its 30.6 m tile fins are over their rate limit.

E4_entry2 at 16 mrad, 15 years, 100 signal events:

	selected	vetoed	let through
signal	100	FP 2.35	TN 97.65
combinatorial	16.8	TP 16.8	FN 10^{-5}
μ helium	4 307	4 307	0
μ PVC walls	128	128	0
μ veto planes	80	80	0
ν helium	0.92	0.29	0.63
ν veto planes	0.06	0.06	0
background	4 532	TP 4 532	**FN 0.63**

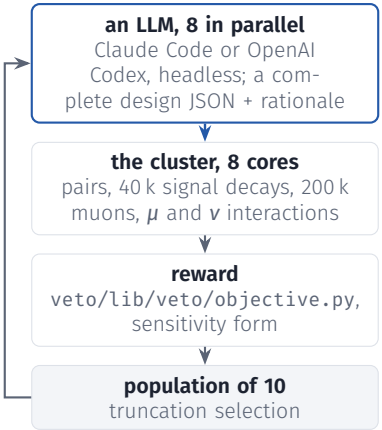
0 muon interactions: fewer than one simulated event of weight 0.52, i.e. below ≈1.2 per 15 years at 90 % CL. What is left is the neutrino floor.

PART 7

Optimisation

An LLM proposes complete designs, the cluster scores them on every background, the best ones breed.

The loop: an LLM proposes, the cluster scores



AlphaEvolve-style, with **designs instead of code**. Each call sees:

- **once**: geometry and limits, how the veto works, technologies, muon flux by side, z and distance from the wall, the objective, the JSON schema
- **per child**: a directive (refine / explore / combine), the parent and 3 inspirations with their reports, the population's rewards
- **per design**: reward ± error, violations, sensitivity, FN per source, FP, cost; per plane: peak rate vs limit, cost, signal it vetoes, pairs it vetoes

At the end the top 5 are rerun with two more seeds: the best **mean** wins.

Backends: `--backend claude` (Claude Code CLI) or `codex` (`codex exec`), both on the user's own login, no tools, no session saved; usage limits are waited out.

> 10 generations, 80 LLM calls, 80 designs: 7 minutes on one node.

The objective: sensitivity

Of **100 signal events**, **FP** are vetoed by random hits; **FN**: background events surviving selection and veto in 15 years, N = combinatorial + μ + ν interactions.

$$\text{sensitivity} = \frac{100 - \text{FP}}{100} \cdot \frac{S_{90}(0)}{S_{90}(N)} \in (0, 1]$$

$S_{90}(N)$: the expected 90 % CL upper limit on the signal with N background events expected. Sensitivity 0.8 means the smallest signal we can exclude is 1/0.8 = 1.25× that of a perfect detector.

N	0	0.06	0.3	0.6	1	3
S_{90}	2.30	2.39	2.71	3.02	3.36	4.55

reward = −100 (1 − sensitivity) − cost − constraints

cost: 2 points per MCHF up to 5 MCHF, 4 above; over **10 MCHF** or any rate limit: infeasible.

Exchange rates, 1 point each: FP +1 (of 100) ≈ FN +0.03 (near N = 0.6) ≈ 0.5 MCHF.

Why not tiers: the first objective ranked background strictly before signal and capped the signal term at 10 % loss. Neutrinos are vetoed only by random hits, like the signal, so the previous run vetoed everything: **FP 74** of 100 for FN 0.06, 7 MCHF.

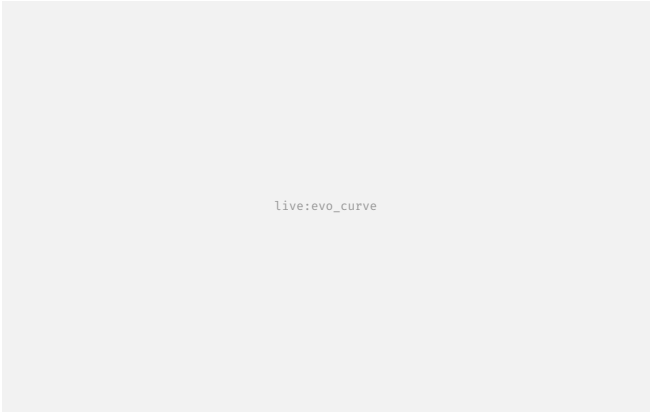
> With the sensitivity, one more FP to remove neutrinos buys back 0.07 points: it never pays.

PART 8

Evolution

Codex evolves the design on the sensitivity objective: 10 generations, 80 designs, 7 minutes.

Evolution



each dot one design (grey: infeasible), orange: the best so far. Hover for names.

OpenAI Codex (gpt-6.1-sol, high effort) on the cluster, 10 generations × 8 children, seeded with the hand-made designs, E1–E3 and E4. Enriched dataset, all backgrounds.

The seeds sit at ≈ −88: without a cone, 76 muon interactions per 15 years survive.

Generation 1 adds a **cone** and **second stations**: −24.7. After that the fins are trimmed centimetre by centimetre: −22.2 from generation 4 on.

> The top five, rerun with two other seeds, agree within 0.03.

EVOLUTION

Watch the design evolve

live:evo3d

live:evo_top

live:evo_step

Each step: a new best design. 3D: planes coloured by muon density. Top view: fins green / orange / red below 50 %, below 100 %, above their rate limit; thick: two stations.

EVOLUTION

Where the designs sit

live:evo_map

Every feasible design: sensitivity against cost, coloured by FP (of 100 signal events); dotted lines of equal reward.

The best designs pile up at **0.5 MCHF and sensitivity 0.79**: none of the 80 found a reason to spend more. Bigger fins only add random hits, and nothing a plane can buy removes the neutrinos.

> Diamonds: the hand-made E3 and E4 and the evolved E5, end to end.

The evolved design, end to end

E5_evolved at 16 mrad, 15 years, **100 signal events**:

	selected	vetoed	let through
signal	100	**FP 0.32**	TN 99.68
combinatorial	16.8	TP 16.8	FN 0.004
μ helium	4 330	4 330	0
μ PVC walls	161	161	0
μ veto planes	80	80	0
ν helium	0.87	0.27	0.60
ν veto planes	0.06	0.06	0
background	4 589	TP 4 588	**FN 0.61**

Rerun with **new** interactions (16 M μ, 7.4 M ν): 0 of 8 337 selected muon interactions survive, i.e. below ≈1.2 at 90 % CL.

	E4_entry2	**E5_evolved**
fins at 30.6 m	0.70 m ×2	**0.38 m** ×2
later fins	36 m: 1.1 m	38.6 m: 0.5 / 0.6 m ×2
cost	0.61 MCHF	**0.54 MCHF**
FP / TN (of 100)	2.35 / 97.65	**0.32 / 99.68**
FN (μ / ν)	0 / 0.63	0 / 0.61
sensitivity	0.739	**0.759**
reward	−27.3	−25.2

both: UBT **×2** (two stations), cone 16 mrad

Shorter entrance fins still catch every parent, and the cone picks up fewer random hits on them. (Evolution: −22.2, with a smaller ν sample.)

> Of 100 signal events 99.7 are kept, next to 0.6 background events: 99.4 % pure. The 0.6 is all neutrinos.

PART 9

Next

What the simulation and the reconstruction still ignore, and what comes next.

NEXT

What is not in yet

Objective and optimisation

- a cautious variant: rate limits at half their value, plane efficiency 0.99
- evolve against independent interaction samples (the evolution sees one fixed draw)

Reconstruction

- tag the neutrino event's own extra products on the fins (wide-angle hadrons downstream of the vertex): the one handle on the neutrino floor
- a dedicated search for the parent's entry hit, instead of a wide cone
- algorithm B: hit-driven tracklets (sees bounces); dead time and pile-up; tilted fins

Physics and data

- statistics: the surviving muon interactions are a handful of simulated events (weight 0.5 each); the first metre beside the walls and the floor have ~10 independent input muons per cell
- epsilon (6 000 seeds) through the chain; an alpha-storage production (≈1 000 seeds) as the next dataset
- photon conversions and secondary interactions (fast chain −5 to −9 % vs Geant4); shield and rock interactions
- specific signal models beyond the two-body decay

NEXT

Summary

Data: our own production keeps every muon category; an input enriched with a Pythia 6 tail gives the high-energy SST1 muons 300× more independent statistics.

Backgrounds: combinatorial pairs, muon interactions (Geant4 + Pythia 6) and neutrino interactions (GENIE), all through one selection (pointing, two tracks) and one veto with a vertex cone.

End to end: the combinatorial background is gone ($< 10^{-2}$); muon interactions need the cone *and* a second station where their parents enter.

Objective: the sensitivity, signal efficiency × $S_{90}(0)/S_{90}(N)$, plus a cost of 2 points per MCHF, 10 MCHF at most. It cannot be gamed by vetoing signal.

Evolution: 80 designs in 7 minutes; the best one, E5, beats the hand-made E4 on signal and cost.

> E5_evolved, 15 years, 100 signal events: FP 0.3, TN 99.7; background TP 4 588, FN 0.6 (all neutrinos). 0.54 MCHF, sensitivity 0.76.